STA442 - Assignment 4 – Influential Factors Amongst 2016 Trump Voters in Wisconsin

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Introduction

The election victory for Donald J. Trump in 2016 came at rather a surprise, as all forecasting has predicted him to lose the election, that his opponent Hillary Clinton to gain an overwhelming victory over him, which this prediction was then overturned completely by the actual result of the election. The purpose of this analysis is to identify the factors that influenced the American voter's decision which brought them to vote for Donald J. Trump in the election in 2016. Here we wish to examine the following, that we question if what are the important demographic factors that was causing a strong spatial pattern in Trump support? Or was there actually little spatial variation with that the Trump voters being evenly distributed in Wisconsin state? Is Trumpism mainly an urban or a rural phenomenon? Does it rather have a relationship with being a racial phenomenon?

Method

The result of an election can be interpreted in a fairly simple way as a yes-no answer to 'voting for president candidate A', which candidate A here we would let it be Donald J. Trump, and it will follow a binomial distribution, with Y_i being the number of votes being casted for Trump.

$$Y_i \sim \text{Binomial}(N_i, p_i)$$

Here N_i presents the total number of votes in the sub county regions *i*, and μ presenting the intercept. We would have X_i presenting all of the predictors, such as Pdens as the population density with logPdens being the log of it, propWhite as the ratio of White raced individuals in the region, and propInd presenting the ratio of indigenous population. We would then have ρ representing the sort or long correlation with ρ_i being the probability of votes being casted for trump, and $\frac{\rho_i}{1-\rho_i}$ would be the odds, and $\log\left(\frac{\rho_i}{1-\rho_i}\right)$being the log odds which is presented below by:

$$\log\left(\frac{\rho_i}{1-\rho_i}\right) = \mu + X_i\beta + U_i$$

And we would give default β prior, which the intercept β_0 would be given a Gaussian prior that mean and precision equals to 0, that the rest of the fixed effects would be given a Gaussian prior with a mean of 0 and precision of 0.001

And as we will be interpreting the data in a geographical way, here we will use a Besag-York-Mollie (BYM) model for U_i, the spatial random effect.

$$U_i \sim \text{BYM}(\sigma^2, \tau^2)$$

$$\begin{bmatrix} U_{S_1} \\ \cdots \\ U_{S_n} \end{bmatrix} \sim MVN \begin{bmatrix} \mu_{S_1} \\ \cdots \\ \mu_{S_n} \end{bmatrix}, \Sigma \Bigg]$$

Which U_i is the spatial random effect, and W_i being an improper Gaussian Markov Random Fields (GMRF) and V_i being an independent noise.

$$U_i = W_i + V_i$$

$$V_i \sim i.i.d.N\ (0, \tau^2)$$

$$W_i | \{W_j; j \neq i\} \sim N(mean\{W_j; j \sim i\}, \frac{\sigma^2}{|j \sim i|})$$

There might be correlation between different geographical locations, which we interpret it as following:

$$Cov(U(s+h), U(s)) = \sigma^2 \rho(\frac{h}{\theta_2}, v)$$

Which we would have *v* being the shape parameter, we can then add on that θ_1 being the marginal standard deviation and θ_2 used above being the spatial proportion that is also the range parameter, with the prior of θ_1 and θ_2 being shown below. And to represent them, we have σ^2 being the spatially structured variance parameter and τ^2 here presents the spatially independent variance:

$$\theta_1 = \sqrt{\sigma^2 + \tau^2}$$

$$\mathrm{pr}(\theta_1 > \log(2.5)) = 0.5$$

$$\theta_2 = \frac{\sigma}{\sqrt{\sigma^2 + \tau^2}}$$

$$\mathrm{pr}(\theta_2 > 0.5) = 0.5$$

Results

Below figure 2 consists of 4 sub-plots, which each presents the demographic features of the counties of Wisconsin. Such as that the first sub-figure a) presents the proportion of the population has voted for Trump, that the closer it is to the red color, the more people in the sub county has voted for Trump, and the further blue the color of the sub county is, the lesser people in the sub county has voted for Trump, while closer it is to a lighter color, that the counties with white colors are the ones having near equal amounts of votes for Trump and Clinton.

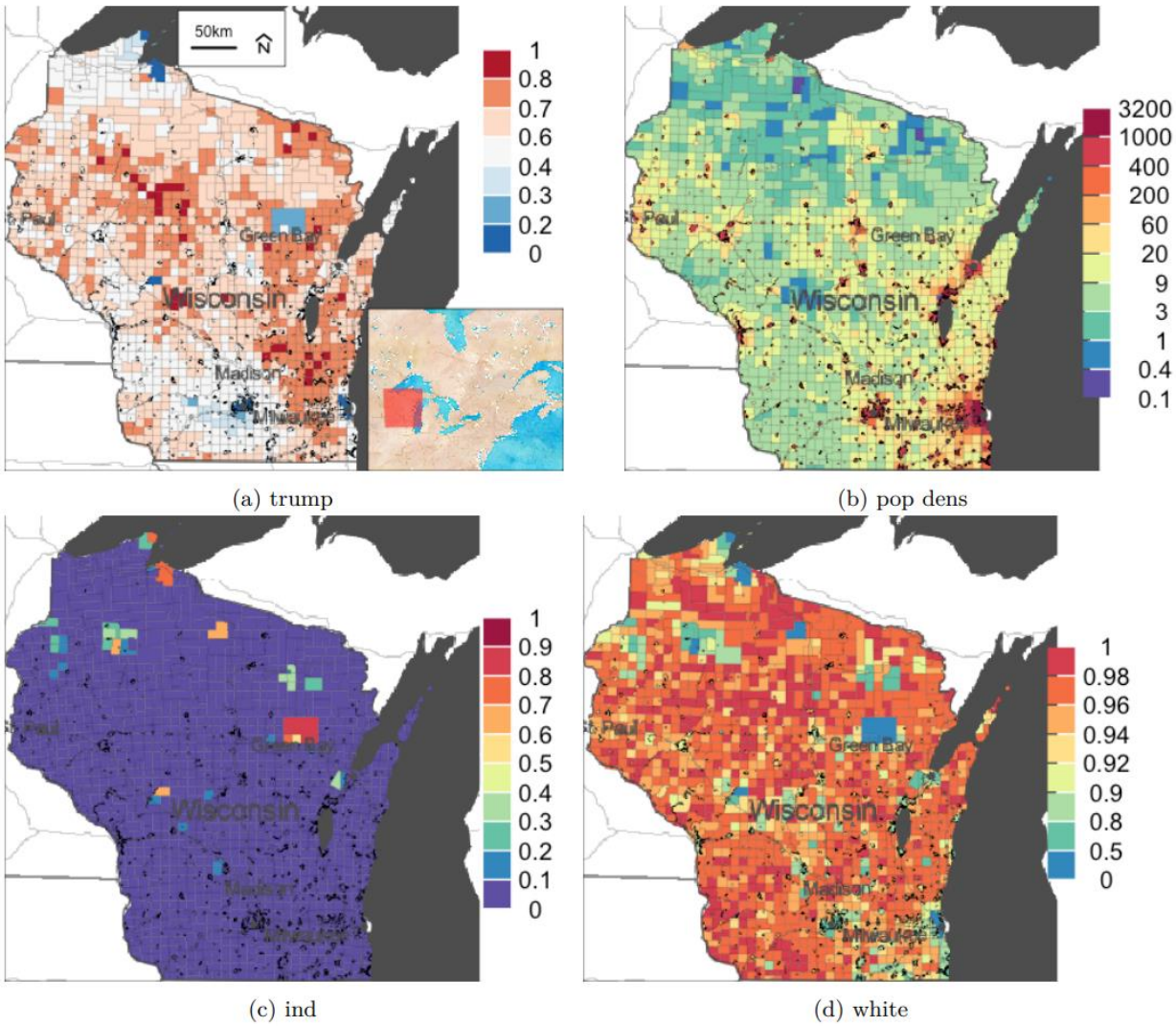


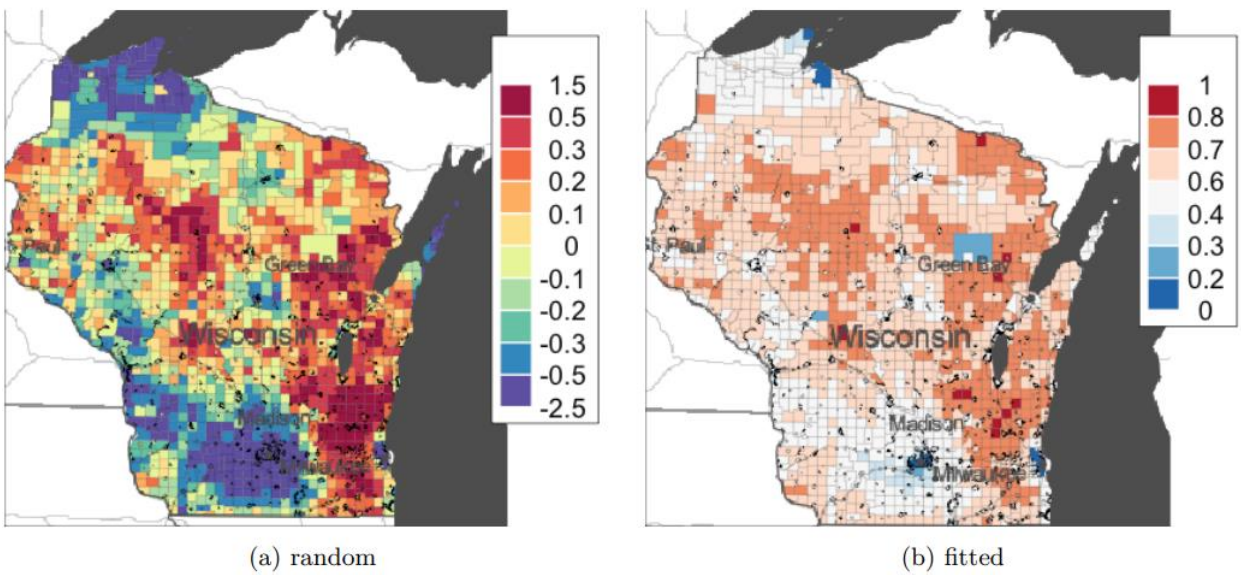
Figure 1 a) trump: proportion of votes for trump in Wisconsin counties
Figure 1 b) pop dens: population density of Wisconsin counties
Figure 1 c) ind: proportion of indigenous population in Wisconsin counties
Figure 1 d) white: proportion of white population in Wisconsin counties

Here to just to make some simpler identifications, we take the counties in red in sub-figure a) to consider that in the entire Wisconsin state, we can see beside a few blue spots, its fairly leaning on

colored red/orange, that the population of the state overall, has more tendency on voting for Trump. And then compare them to in sub-figure b) we see that the heavily populated counties of Milwaukee, Madison and Green Bay, that on the same location has a blue color in sub-figure a), which it implies that the urbanized populations of Wisconsin are lesser likely to vote for Trump, and the yellow and blue sub counties which are the rural regions, have a higher tendency for casting their vote for Trump.

And to sub-figure c), that we see concentrated indigenous communities north-west of Green Bay, around Keshena, and on figure a), it showed a large box blue color, which implies that the native population are a lot lesser likely to vote for Trump.

Lastly, we take sub-figure d), which are the concentration of the white raced individuals in Wisconsin counties, as it is fairly messy, we will take the counties east of Green Bay, that have relatively high proportion of white raced individuals, that compare to figure 1, that these counties with high proportion of white raced individuals does have somewhat higher tendency on having their votes casted for Trump, and as there are more sub counties with more white populations in Wisconsin, it make sense of the evenly distribution of voters through out Wisconsin state. Thus, from the results above, we can pretty much conclude that a fairly strong spatial pattern is indeed present on the result of the election.



(a) random (b) fitted

Figure 2 a) posterior spatial random effect in each sub county
Figure 2 b) predicted probability of voting for trump in each sub county

Figure 2 a) from above shows the spatial random effect in each sub county, which for which that we see all of the spectrums from being positive to negative are all present, which compares to figure 1 a), that our model should do a fairly good job on predicting.

And from figure 2 b), we can tell that that our model does a great job on predicting the probability on which counties are likely to have more votes for Trump and which are likely to have lesser, such that compare to figure 1a), the orange and deeper colored, and lighter colored and blue colored spots approximately all matches.

However, as above are judgements made through our human eyes, which are not accurate and are very likely to contain bias and error as the graphs contain way too much detailed information to be processed by just the human eye and brain. So here we introduce the quantile table of odds ratio below for the random effects, for a better visualization of our data.

	0.5quant (median)	*0.025 quant*	*0.975quant*
(intercept)	0.56963	0.4372894	0.7432372
logPdens	0.9221476	0.9141231	0.9302332
propWhite	4.132118	3.165813	5.382054
propInd	0.4541036	0.3216472	0.6400045
Sd	1.374789	1.355527	1.397186
propSpatial	2.612114	2.502149	2.68025

Figure 3: Quantile table for odds ratios of probability on Trump votes, standard deviation, and spatial proportion for random effects

So, we would start interpreting with above quantile table of odds ratio for random effects. As this table is the results after we have taken the exponentials, which that as $e^0 = 1$, we would be comparing the results in this table with 1 rather than 0 to judge if it is significant. And here we care most about the 3 predictors, which are the blocks highlighted in Red that consists of logPdens, propWhite and propInd, the factors about the demographics of the Wisconsin counties.

For logPdens, from the table we can tell that when $\log X_2 - \log X_1 = 1$, which that as the population density increase by 1 percent, that the odds ratio would be to 0.9221476 times or lowed by 0.0778524%, and that $\log X_2 - \log X_1 = 1$ is the same as $\log \frac{X_2}{X_1} = 1$, and thus $\frac{X_2}{X_1} = e \approx 2.718$, and that implies that when the population density increases to about 2.718 times, then the odds would decrease by approximately $\approx 10\%$

Here we see that the quantile for propInd does not contain 1 in it, which would imply that it is not so significantly influential, and the value of their 0.975 quantile is lower than 1. Here we interpret the median that it would imply that its influences are actually negative for casting their votes for Trump, that as the proportion of Indigenous people or population density increases in a population of the sub county, their chance on voting for Trump is lowed by 0.5458964%.

And lastly propWhite does contain 1 in all the quantiles, and all are larger than 1, which implies that as the proportion of White raced individuals increase by 1%, the probability on casting their votes for Trump would become 4.132118 times higher, that counties being predominantly white would have a significantly positive influence on voting for Donald J. Trump.

Summery

This study is conducted with the purpose of to obtain further insight of if the demographic factors had an impact on the 2016 presidential election of the United States, that we wish to examine what are the important demographic factors that was causing a strong spatial pattern in support for Trump, or if there was little spatial variation with that the Trump voters being evenly distributed in Wisconsin state, and the hypothesis that Trumpism being mainly an urban or a rural phenomenon, and if it rather have a relationship with being a racial phenomenon.

And through the data of the election and our generated model, both graphicly and numerically, we can conclude first on our questioning with that Trumpism is both a rural phenomenon and a racial phenomenon, as those in a rural region are more likely to vote for Trump and those in urban regions such as Milwaukee with high population density are lesser likely to vote for Trump, while the lesser dense rural regions are more likely to vote for Trump, with relationship to the white race and indigenous, we can see that the sub counties that are more populated with indigenous population are very unlikely to vote for Trump, while the mostly white sub counties are more likely to vote for him. And a spatial pattern of fair strength is present for the support of Trump in the 2016 presidential election.